Module 5 - Assignment 3 R Shiny

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ALY6070: Communication & Visualization for Data Analytics

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Introduction

The electric vehicle (EV) market has experienced rapid growth over the past decade, driven by environmental concerns, government incentives, and advancements in battery technology. Understanding the patterns behind EV adoption - which brands dominate, how range and pricing have evolved, and where registrations are concentrated is essential for policymakers, automakers, and consumers alike.

This report presents an interactive Shiny dashboard built in R, analyzing the Electric Vehicle Population dataset sourced from the Washington State Department of Licensing via Kaggle. The dataset captures Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) currently registered in Washington State. The goal of this assignment is to explore this dataset through interactive visualizations, uncover meaningful trends in EV adoption, and demonstrate how storytelling with R Shiny can turn raw registration data into actionable insights.

The analysis covers several dimensions including brand and model popularity, the growth of EV registrations over time, differences between BEV and PHEV vehicles, geographic distribution across counties and cities, electric range and pricing trends, and clean fuel eligibility patterns. Two calculated fields - Price Tier and Range per Dollar - were created to add deeper analytical value beyond the raw data.

Analysis

The dataset contains approximately 200,000 rows, where each row represents a single registered electric vehicle in Washington State. The data spans model years from the early 2000s through 2024, capturing over two decades of EV registrations.

The dataset includes 17 original variables, which can be categorized as follows:

Categorical Variables: County, City, State, Brand (Make), Model, Electric Vehicle Type (BEV or PHEV), Clean Alternative Fuel Vehicle (CAFV) Eligibility, Electric Utility, and Vehicle Location.

Discrete Variables: Model Year, Postal Code, Legislative District, DOL Vehicle ID, and Census Tract.

Continuous Variables: Electric Range (in miles) and Base MSRP (in US dollars).

In addition to the original variables, two **calculated fields** were derived for this analysis.

Price Tier categorizes each vehicle into Budget (under $25K), Mid-Range ($25K to $45K), Premium ($45K to $70K), or Luxury (over $70K) based on the Base MSRP.

Range per Dollar measures efficiency as miles of electric range per $1,000 of MSRP, where a higher value indicates better value for money.

Target Audience

The primary audience for this dashboard includes government agencies and policymakers evaluating the effectiveness of EV incentives and CAFV programs, automotive industry analysts tracking brand performance and market trends, electric utility companies planning infrastructure for EV charging, urban planners studying geographic adoption patterns, and consumers comparing brands based on range, pricing, and value.

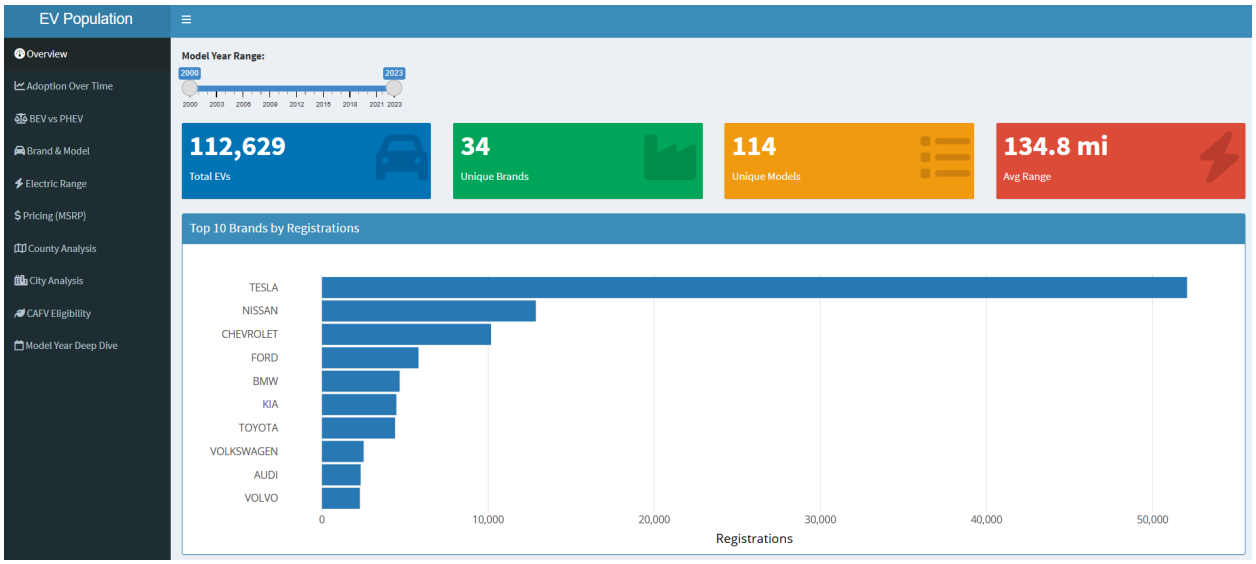
Dashboard Visualizations

The following sections present the visualizations built into the Shiny dashboard. Each tab addresses a specific analytical question, and every graph is accompanied by the question it aims to answer, a labeled figure, and a description of the findings.

Tab 1: Overview

Question: What does the overall EV landscape look like in Washington State, and which brands lead in registrations?

Figure 1: Overview Dashboard - KPI Cards and Top 10 Brands by Registrations



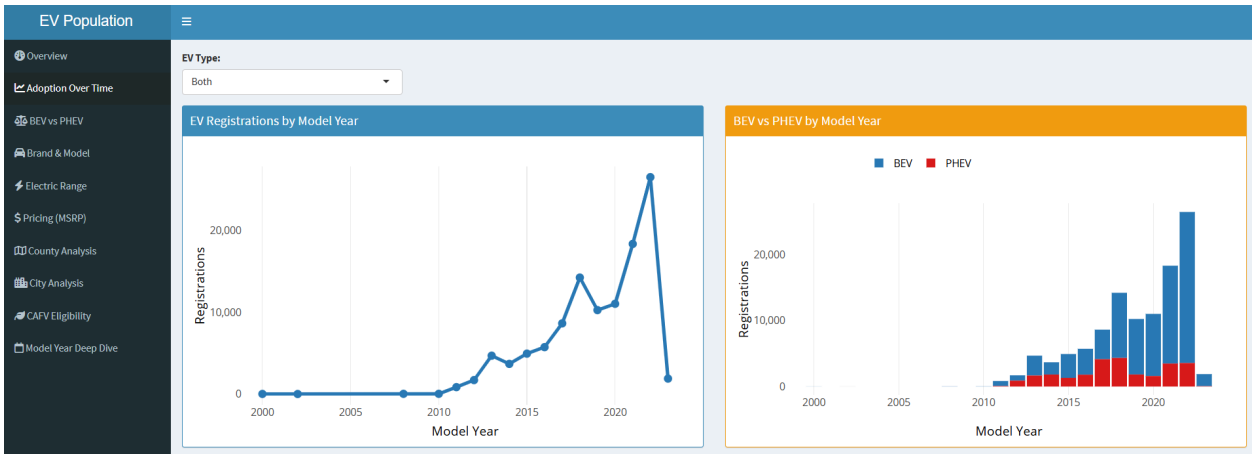
The Overview tab provides a high-level snapshot of the dataset through four KPI value boxes and a summary bar chart. With the slider set to the full range (2000–2023), the dashboard reveals 112,629 total registered EVs across 34 unique brands and 114 unique models, with an average electric range of 134.8 miles. The horizontal bar chart ranks the top 10 brands by registration count. Tesla dominates with over 50,000 registrations, far exceeding Nissan and Chevrolet which follow in second and third place respectively. The remaining brands — Ford, BMW, Kia, Toyota, Volkswagen, Audi, and Volvo — each hold a comparatively smaller share of

the market. The model year slider allows users to dynamically filter the view, and when narrowed to earlier years, the KPIs reveal how sparse EV adoption was before 2010.

Tab 2: Adoption Over Time

Question: How has EV adoption grown over the years, and what is the trend between BEV and PHEV registrations?

Figure 2: EV Registrations by Model Year and BEV vs PHEV Breakdown

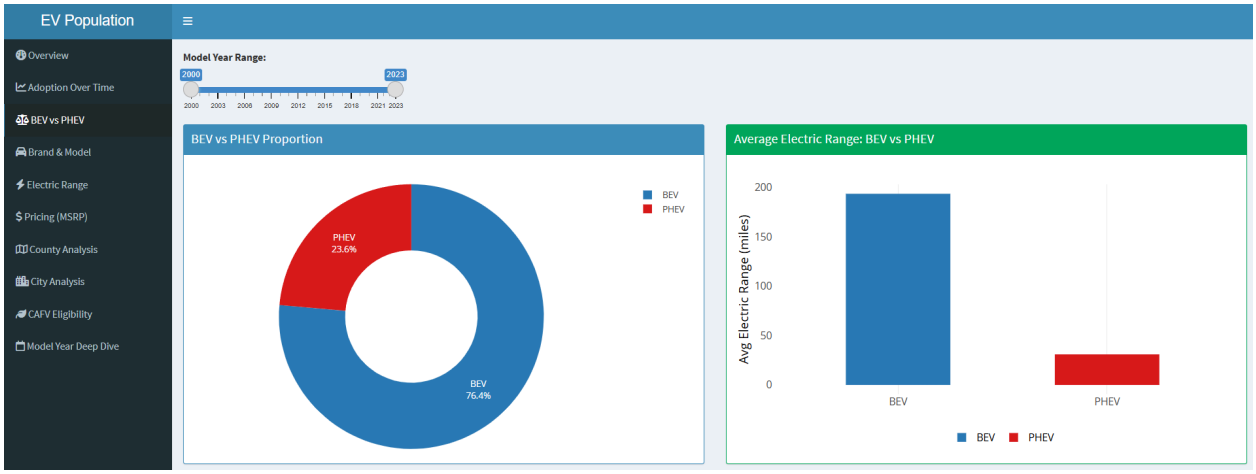


The first graph is a line chart showing total EV registrations by model year. The trend is relatively flat from 2000 through 2010, after which registrations begin climbing steadily. A sharp acceleration occurs around 2018, with registrations peaking around 2022–2023 at over 25,000 vehicles in a single model year. The slight dip at the end likely reflects incomplete data for the most recent year. The second graph is a stacked bar chart breaking down registrations into BEV (blue) and PHEV (red) categories. BEVs consistently outnumber PHEVs and the gap has widened dramatically in recent years, indicating a strong and growing consumer preference for fully electric vehicles over plug-in hybrids. An EV Type dropdown filter allows the user to isolate either type.

Tab 3: BEV vs PHEV

Question: What is the overall proportion of BEVs to PHEVs, and how do their electric ranges compare?

Figure 3: BEV vs PHEV Proportion and Average Electric Range Comparison

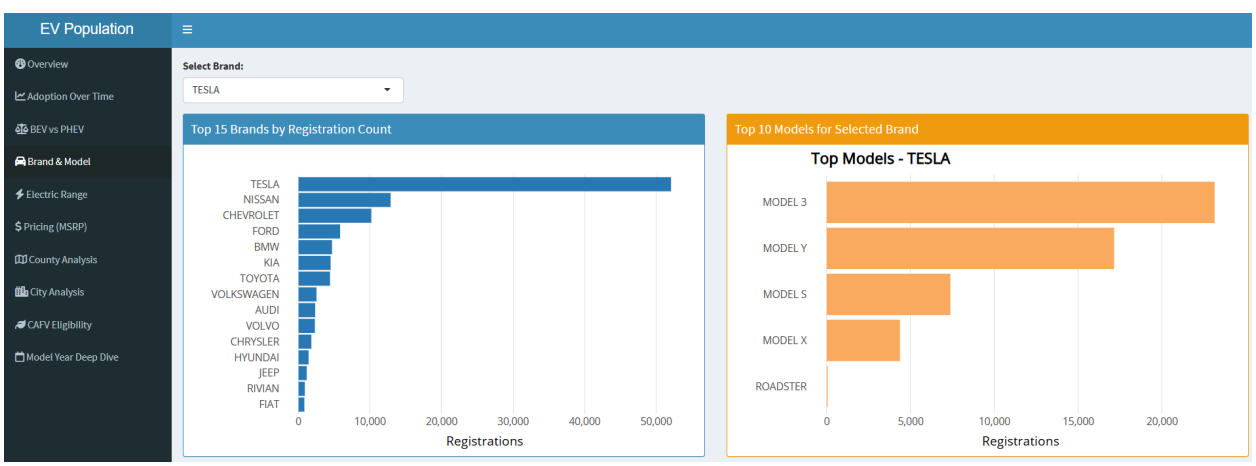


The donut chart shows that BEVs account for 76.4% of all registered EVs while PHEVs make up the remaining 23.6%. This confirms that the majority of EV owners in Washington State prefer fully electric vehicles. The bar chart on the right compares the average electric range between the two types. BEVs have an average range of approximately 190 miles, while PHEVs average significantly lower at around 30 miles. This large gap is expected since PHEVs rely on a combination of electric and gasoline power and are designed for shorter electric-only trips. The model year slider enables users to observe how this proportion has shifted over different time periods.

Tab 4: Brand & Model

Question: Which brands dominate the EV market, and what are the most popular models within each brand?

Figure 4: Top 15 Brands and Top Models for Tesla



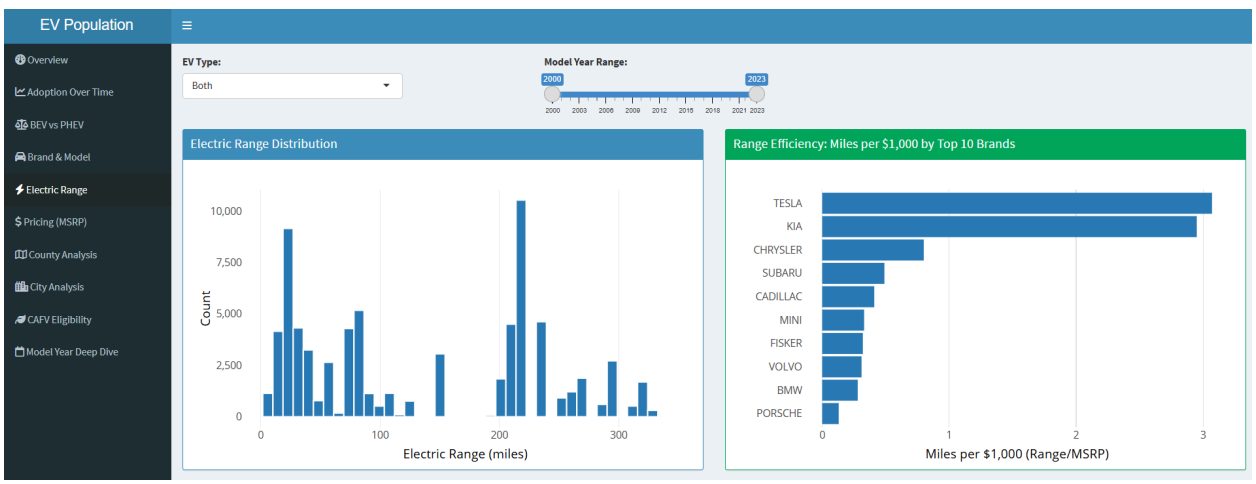
The left chart expands the brand view to the top 15, where Tesla's dominance becomes even more pronounced - its bar extends well beyond 50,000 registrations while the next closest

brand, Nissan, sits around 10,000. The long tail of smaller brands like Chrysler, Hyundai, Jeep, Rivian, and Fiat shows the market is highly concentrated at the top. The right chart is dynamic and updates based on the brand dropdown. With Tesla selected, the Model 3 leads as the most registered model with over 20,000 units, followed closely by the Model Y. The Model S and Model X trail behind, with the Roadster having minimal registrations. This interactive feature allows exploration of any brand's model breakdown.

Tab 5: Electric Range

Question: How is electric range distributed across EVs, and which brands offer the best range per dollar?

Figure 5: Electric Range Distribution and Range Efficiency by Top 10 Brands

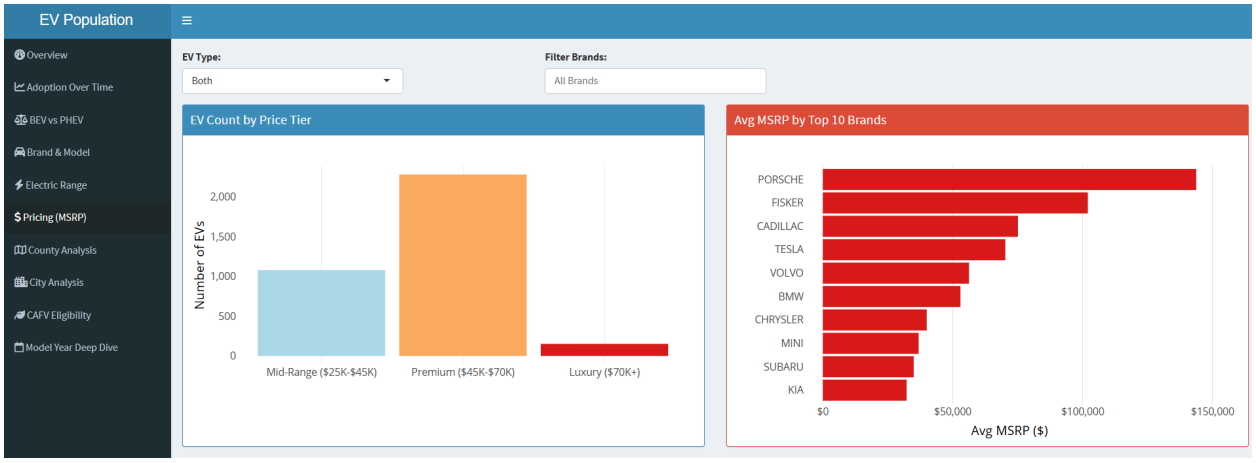


The histogram on the left reveals a bimodal distribution of electric range. There is a large cluster of vehicles in the 0–50 mile range, which corresponds to PHEVs and older BEVs with limited battery capacity. A second prominent peak appears around the 200–300 mile range, representing modern BEVs with advanced battery technology. The gap in between reflects the technological transition period. The right chart uses the calculated field Range per Dollar (miles per $1,000 of MSRP) to measure efficiency. Tesla and Kia lead significantly, offering the highest range relative to price. Brands like Porsche and BMW appear at the bottom, indicating that while they offer capable EVs, the range-to-cost ratio is lower — typical of luxury positioning. The EV Type and Model Year filters allow further segmentation.

Tab 6: Pricing (MSRP)

Question: How are EVs distributed across price tiers, and which brands have the highest average pricing?

Figure 6: EV Count by Price Tier and Average MSRP by Top 10 Brands

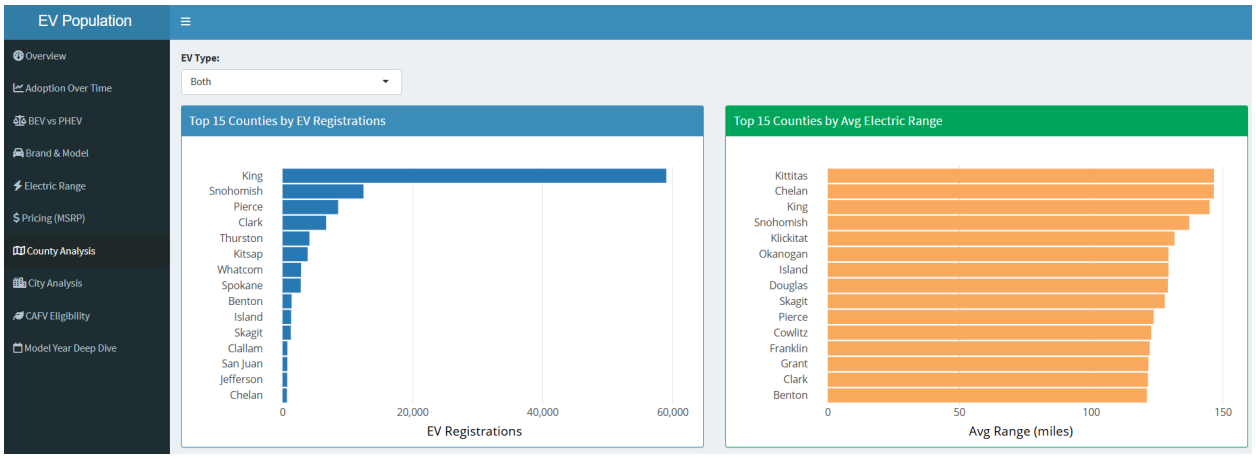


The left chart uses the calculated Price Tier field to categorize vehicles into three segments. The Premium tier ($45K–$70K) dominates with over 2,200 vehicles, followed by Mid-Range ($25K–$45K) with around 1,000. The Luxury tier ($70K+) has a relatively small count, indicating that most registered EVs fall in the mid-to-premium price range. The absence of a Budget tier (under $25K) in the chart suggests that truly affordable EVs are rare in this dataset. The right chart ranks the top 10 brands by average MSRP. Porsche leads at approximately $150,000, followed by Fisker and Cadillac. Tesla sits in the mid-range around $50,000, which combined with its high range makes it a strong value proposition. The EV Type and Brand filters allow users to drill into specific segments.

Tab 7: County Analysis

Question: Which counties have the highest EV adoption, and where are longer-range EVs concentrated?

Figure 7: Top 15 Counties by Registrations and Average Electric Range



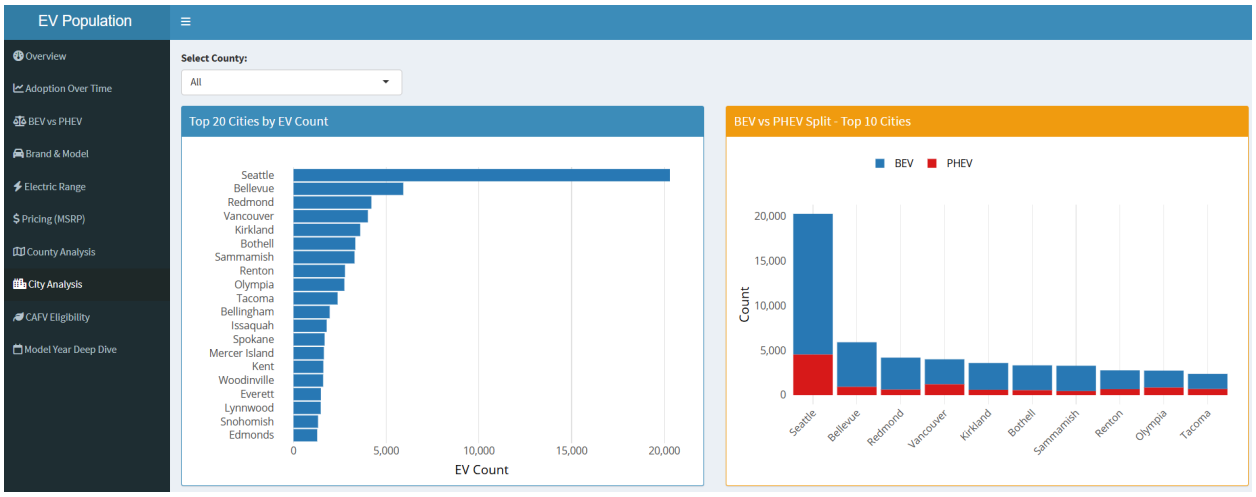
The left chart shows King County leading by a massive margin with over 60,000 registrations - more than all other counties combined. Snohomish and Pierce follow in distant second and third

positions. This heavy concentration in King County aligns with the Seattle metropolitan area being a tech-forward, environmentally conscious region with higher income levels. The right chart shifts the focus to average electric range by county. Interestingly, Kittitas and Chelan counties — which are smaller and more rural — top the list for average range, suggesting that EV owners in these areas tend to purchase longer-range vehicles, likely due to the greater distances they need to cover. The EV Type filter allows users to compare geographic patterns between BEVs and PHEVs.

Tab 8: City Analysis

Question: Which cities lead in EV adoption, and how does the BEV vs PHEV split vary across cities?

Figure 8: Top 20 Cities by EV Count and BEV vs PHEV Split

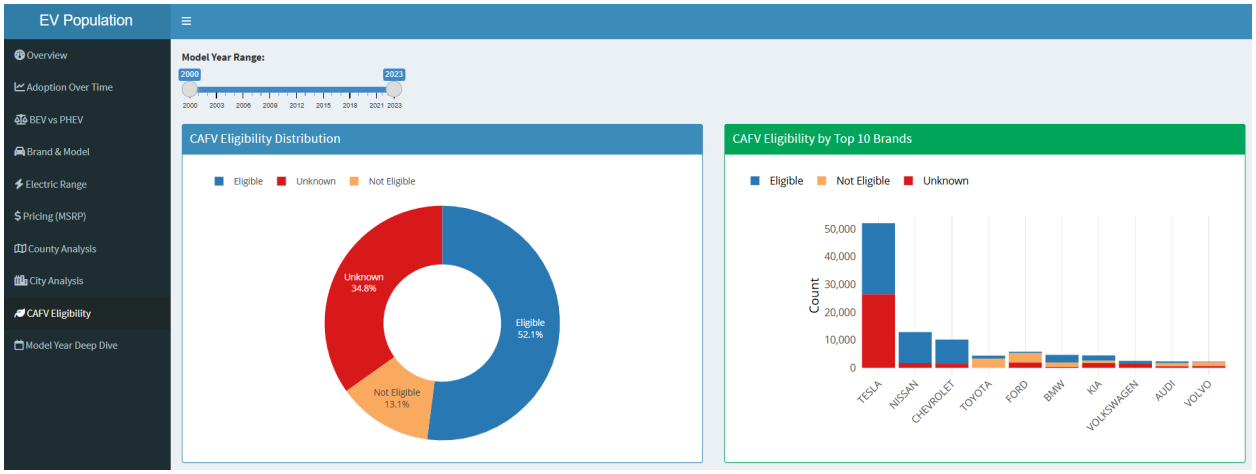


The left chart ranks the top 20 cities, with Seattle leading at over 22,000 registrations. Bellevue and Redmond follow, both being major tech hubs within the Greater Seattle area. Vancouver, Kirkland, and Bothell round out the top six. The chart shows a steep drop-off after the first few cities, indicating that EV adoption is heavily urban-concentrated. The stacked bar chart on the right breaks down the top 10 cities by BEV and PHEV. Across all cities, BEVs (blue) dominate, but the PHEV share (red) is relatively more visible in Seattle and Bellevue compared to smaller cities. The county dropdown filter enables city-level exploration within any specific county.

Tab 9: CAFV Eligibility

Question: What proportion of EVs qualify for clean alternative fuel incentives, and how does eligibility vary by brand?

Figure 9: CAFV Eligibility Distribution and Eligibility by Top 10 Brands

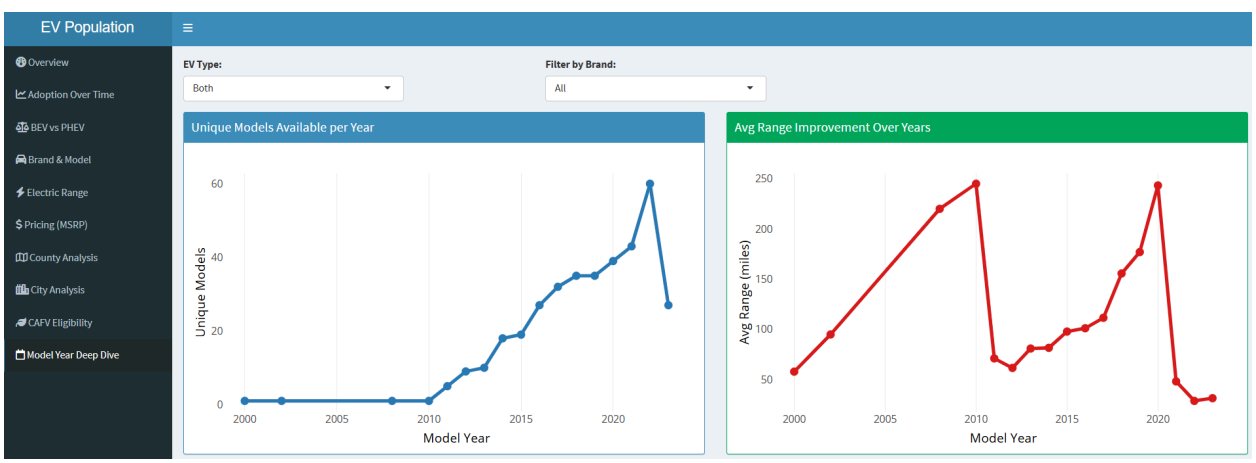


The donut chart reveals that 52.1% of registered EVs are classified as Clean Alternative Fuel Vehicle Eligible, while 34.8% have an Unknown eligibility status and 13.1% are Not Eligible due to low battery range. The significant Unknown portion suggests gaps in data reporting or vehicles that have not yet been assessed. The stacked bar chart on the right breaks this down by brand. Tesla shows a large proportion of eligible vehicles alongside a notable red (Unknown) segment. Nissan and Chevrolet also have mixed eligibility profiles. Smaller brands like BMW, Kia, and Ford show varying proportions of eligible, unknown, and ineligible vehicles. This tab is particularly valuable for policymakers assessing the reach of clean fuel incentive programs.

Tab 10: Model Year Deep Dive

Question: How has the variety of EV models and average range evolved over model years?

Figure 10: Unique Models per Year and Average Range Improvement Over Years



The left line chart tracks the number of unique EV models available per model year. From 2000 to 2010, the market offered fewer than 5 models. After 2010, model variety expanded rapidly, peaking at over 60 unique models around 2022–2023 before a slight dip in the most recent year.

This growth reflects increasing competition and investment in the EV space from both established automakers and new entrants. The right line chart shows the average electric range over time. Starting around 50 miles in 2000, the range climbed to approximately 250 miles by 2018–2019. However, the chart shows some fluctuation in recent years, with a dip visible around 2022–2023. This could be explained by the influx of shorter-range PHEVs and budget-oriented BEVs entering the market alongside longer-range premium models, pulling the average down. The EV Type and Brand filters allow users to isolate these trends for specific segments.

Further Exploration

While the dashboard covers a wide range of analytical dimensions, several additional questions could be explored with further analysis.

How does EV adoption compare across different states if a national dataset were available? A choropleth map visualization could effectively show adoption density across the entire United States, highlighting regional disparities.

What is the relationship between household income levels and EV registrations at the county level? Joining this dataset with census income data and creating scatter plots or heatmaps could reveal whether higher-income areas correlate with greater EV adoption.

How do seasonal patterns affect EV registrations? A time series analysis with monthly granularity - if registration dates were available - could uncover whether EV purchases peak during certain months, potentially tied to tax incentive deadlines or new model releases.

What is the resale value trajectory of different EV brands? If secondary market pricing data were integrated, a line chart tracking depreciation by brand and model year could help consumers make more informed purchasing decisions.

How does charging infrastructure availability influence adoption at the city level? Overlaying charging station data with registration counts through a combined bar and line chart could show whether cities with more charging stations also have higher EV adoption rates.

Conclusion

This analysis of the Washington State Electric Vehicle Population dataset reveals a rapidly evolving market with clear trends. EV adoption has grown exponentially, particularly after 2018, with Tesla commanding a dominant market share across all metrics. BEVs account for over three-quarters of all registrations and continue to outpace PHEVs, signaling a clear consumer shift toward fully electric vehicles. Geographically, King County and the Seattle metropolitan area represent the epicenter of EV adoption, though rural counties show interesting patterns in range preferences. The pricing analysis highlights that most EVs fall in the Premium tier, with Budget options remaining scarce - an important consideration for expanding EV access to

broader demographics. The calculated Range per Dollar metric reveals that Tesla and Kia offer the strongest value in terms of electric range relative to cost, while luxury brands prioritize performance over efficiency.

The interactive Shiny dashboard built for this project demonstrates how R can be used to transform a static dataset into a dynamic storytelling tool. With 10 tabs, 20 interactive visualizations, and multiple filters, the dashboard enables users to explore the data from multiple angles and draw their own conclusions - making it a practical resource for analysts, policymakers, and consumers interested in the electric vehicle landscape.

References

Kakkar, R. (n.d.). Electric Vehicle Population Data. Kaggle.

https://www.kaggle.com/datasets/ratikkakkar/electric-vehicle-population-data

Shiny Dashboard. (n.d.). rstudio.github.io. Retrieved March 20, 2026, from

https://rstudio.github.io/shinydashboard/